

Spotless: Grid-Free Deconvolution for Radio Interferometry

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Abstract

Spotless is a point-source deconvolution algorithm for radio interferometry that operates directly in visibility space, without gridding. By exploiting Parseval’s theorem, the total energy in the image can be computed from the visibilities without performing a Fourier transform. Spotless builds a model of the sky as a sum of point sources by minimising the residual energy. This approach handles non-coplanar baselines and all-sky fields of view without requiring w -projection or other wide-field corrections. Two variants are presented: sequential spotless (analogous to CLEAN) and MultiSpotless, which jointly optimises all model components.

1 Introduction

Radio interferometers measure the complex visibility function $V(u, v, w)$, which is related to the sky brightness distribution $I(l, m)$ through the van Cittert–Zernike theorem [1]:

$$V(u, v, w) = \iint \frac{I(l, m)}{\sqrt{1 - l^2 - m^2}} e^{-2\pi j[ul + vm + w(\sqrt{1 - l^2 - m^2} - 1)]} dl dm. \quad (1)$$

Here (l, m, n) are direction cosines relative to the phase centre, and (u, v, w) are baseline coordinates in wavelengths.

Traditional imaging pipelines grid the visibilities onto a regular (u, v) -plane and apply a Fast Fourier Transform (FFT) to produce a dirty image. The CLEAN algorithm [2] then deconvolves the dirty image by iteratively subtracting scaled copies of the dirty beam. Variants such as the Cotton–Schwab algorithm [3] perform the subtraction in visibility space to reduce aliasing artefacts.

However, gridding introduces several complications: w -terms cause distortions for non-coplanar arrays; all-sky imaging requires careful treatment of the horizon; and wide-field imaging suffers from direction-dependent effects that are difficult to model on a regular grid.

Spotless avoids gridding entirely by working directly with the visibilities. The key insight is Parseval’s theorem: the total energy (power) of an image can be computed from the visibilities without performing a Fourier transform. This enables a model-fitting approach where point sources are located and characterised by minimising the residual visibility energy.

2 Theory

2.1 Point-Source Visibility

A point source at direction cosines (l, m) with brightness A produces a visibility:

$$V_P(u, v, w) = A^2 e^{-2\pi j[ul + vm + w(n-1)]}, \quad (2)$$

where $n = \sqrt{1 - l^2 - m^2}$. In the Spotless implementation, the source position is parameterised in elevation–azimuth coordinates (θ, ϕ) rather than (l, m) , with the conversion:

$$l = \sin \phi \cos \theta, \quad (3)$$

$$m = \cos \phi \cos \theta, \quad (4)$$

$$n = \sin \theta. \quad (5)$$

2.2 Energy in Visibility Space

For discrete visibilities v_k measured at M baseline points, Parseval’s theorem for the Discrete Fourier Transform states:

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2, \quad (6)$$

where x_n is the image and X_k is its Fourier transform. Since the visibilities are samples of the Fourier transform of the sky brightness, the total energy $E(V)$ in the image is proportional to the sum of squared visibility amplitudes:

$$E(V) \propto \sum_{k=0}^{M-1} v_k v_k^*. \quad (7)$$

No gridding or inverse Fourier transform is required to compute this quantity—it is evaluated directly from the visibility data.

2.3 Sequential Spotless

The sequential Spotless algorithm builds a model of N point sources by iteratively finding and subtracting the brightest remaining source. Beginning with the measured visibilities $V^{(0)} = V$, each step $i = 0, 1, \dots$ solves:

$$(A_i, \theta_i, \phi_i) = \arg \min_{A, \theta, \phi} E(V^{(i)} - A^2 V_P(\theta, \phi)), \quad (8)$$

where $V_P(\theta, \phi)$ is the unit-amplitude point-source visibility from equation (2). The source is added to the model and subtracted from the residual:

$$\mathcal{M} \leftarrow \mathcal{M} \cup \{(A_i, \theta_i, \phi_i)\}, \quad (9)$$

$$V^{(i+1)} = V^{(i)} - A_i^2 V_P(\theta_i, \phi_i). \quad (10)$$

The algorithm terminates when the residual energy stops decreasing, i.e., when $E(V^{(i+1)}) \geq E(V^{(i)})$.

The cost function in equation (8) is a smooth function of three parameters (A, θ, ϕ) . The optimisation is performed using the L-BFGS-B algorithm [4], a quasi-Newton method that supports bound constraints and uses a limited-memory Hessian approximation. An initial guess is obtained by imaging the current residual onto a HEALPix sphere [6] and locating the brightest pixel.

2.4 MultiSpotless

MultiSpotless extends the sequential approach by jointly re-optimising all model components at each step. After i sources have been found, the $(i + 1)$ -th step solves the full $3(i + 1)$ -dimensional problem:

$$\arg \min_{\{A_k, \theta_k, \phi_k\}_{k=0}^i} E\left(V - \sum_{k=0}^i A_k^2 V_P(\theta_k, \phi_k)\right). \quad (11)$$

This allows existing sources to be refined as new ones are added, potentially yielding a more accurate model at the cost of increased computational complexity. Because the full parameter space mixes amplitudes and angles at very different scales, MultiSpotless employs the Nelder–Mead simplex algorithm [5] with adaptive parameters, which is derivative-free and robust to poor scaling.

3 Algorithm

Algorithm 1 Sequential Spotless

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1: Input: Visibilities  $V$ , field-of-view sphere  $\mathcal{S}$ 
2:  $\mathcal{M} \leftarrow \emptyset$ 
3:  $V_{\text{res}} \leftarrow V$ 
4:  $E_0 \leftarrow \infty$ 
5: repeat
6:   Image  $V_{\text{res}}$  onto  $\mathcal{S}$ 
7:    $(A_0, \theta_0, \phi_0) \leftarrow \text{peak of image}$ 
8:    $x_0 \leftarrow [A_0, \theta_0, \phi_0]$ 
9:    $x^* \leftarrow \arg \min_x E(V_{\text{res}} - V_P(x))$ 
10:   $(A, \theta, \phi) \leftarrow x^*$ 
11:   $E_1 \leftarrow E(V_{\text{res}} - A^2 V_P(\theta, \phi))$ 
12:  if  $E_1 \geq E_0$  then break
13:  end if
14:   $\mathcal{M} \leftarrow \mathcal{M} \cup \{(A, \theta, \phi)\}$ 
15:   $V_{\text{res}} \leftarrow V_{\text{res}} - A^2 V_P(\theta, \phi)$ 
16:   $E_0 \leftarrow E_1$ 
17: until max iterations
18: return  $\mathcal{M}, V_{\text{res}}$ 

```

The MultiSpotless variant replaces the single-source optimisation (line 8) with a joint optimisation over all model parameters as in equation (11).

4 Implementation

Spotless is implemented in Python and distributed as the `spotless` package. Key design features include:

- **Grid-free operation.** All computations are performed directly on the visibility data; no gridding or degridding steps are required.
- **All-sky support.** The field of view is represented as a HEALPix sphere, enabling natural handling of wide-field and all-sky observations.
- **Non-coplanar baselines.** The full (u, v, w) baseline coordinates are used in the point-source visibility calculation (equation 2), avoiding the need for w -projection.
- **Plug-and-play data sources.** Spotless accepts visibilities from a TART telescope API, from JSON snapshot files, or from CASA measurement sets (via `disko`).
- **Multiple output formats.** Reconstructed images can be exported as FITS, PNG, SVG, PDF, or HDF5.

The initial peak search and image reconstruction use the HEALPix pixelisation scheme. Beam smoothing during reconstruction is performed with the `healpy` spherical harmonic transform.

5 Results

Figure 1 shows a comparison between a dirty image and a Spotless reconstruction from TART telescope data. Spotless produces a clean point-source model with substantially reduced sidelobe artefacts.

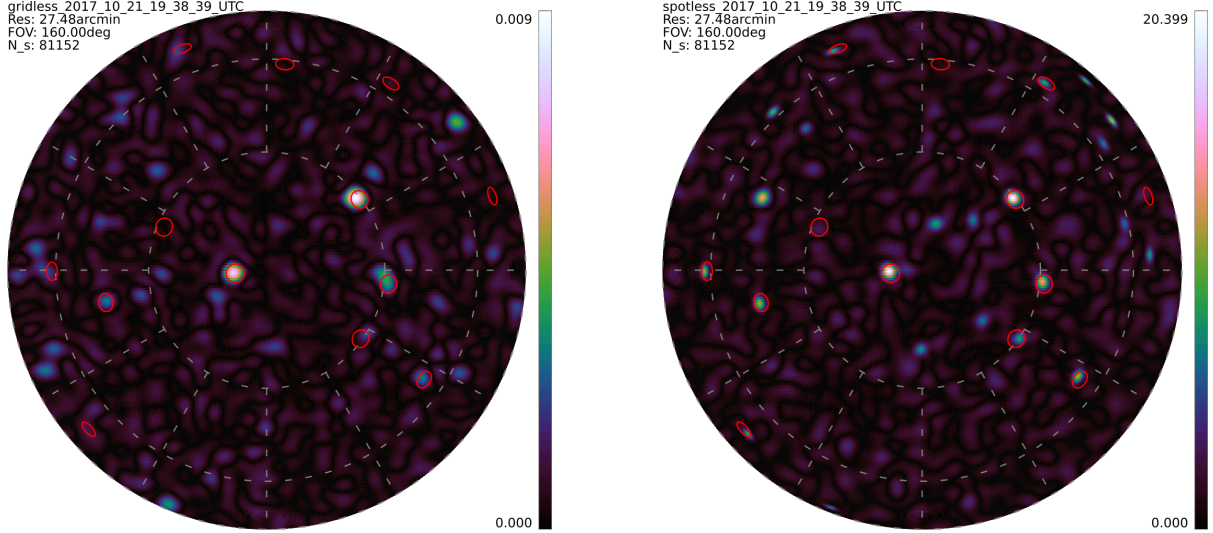


Figure 1: Left: dirty (gridless) image. Right: Spotless reconstruction showing the deconvolved point-source model.

6 Discussion

6.1 Advantages over CLEAN

Spotless offers several advantages over traditional CLEAN-based methods:

1. No gridding artefacts—the model is fit directly to the visibility data.
2. No requirement for a regular (u, v) sampling.
3. Natural handling of non-coplanar baselines via the full (u, v, w) phase term.
4. All-sky fields of view are handled without special treatment.
5. The model is a compact parametric representation (list of point sources) rather than a pixel map.

6.2 Limitations and Future Work

- **Point-source assumption.** Spotless models the sky as a collection of point sources. Extended emission is not well represented. Future work could incorporate Gaussian or shapelet source models.
- **Antenna beam.** The current implementation assumes an isotropic hemispherical antenna response. Incorporating a direction-dependent beam model would improve accuracy, particularly at low elevations.
- **Uncertainty quantification.** Running a Markov Chain Monte Carlo (MCMC) sampler on the MultiSpotless model would provide uncertainty estimates for source parameters and a natural stopping criterion based on posterior precision.
- **Computational cost.** MultiSpotless scales as $\mathcal{O}(N^2)$ in the number of sources due to the joint optimisation. Block-coordinate descent or alternating optimisation could reduce this cost.

7 Conclusion

Spotless provides a grid-free approach to radio interferometric deconvolution by exploiting Parseval's theorem to compute image energy directly from visibilities. By fitting a point-source model through energy minimisation, it avoids the gridding and degridding steps required by traditional imaging pipelines. The sequential variant offers speed comparable to CLEAN, while MultiSpotless provides improved accuracy through joint optimisation. The method is particularly well-suited to all-sky imaging with non-coplanar arrays.

References

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